FCT SERVICE NOTE
SN07061815a

Class C information is Property of GE.

This short-term informational note is invalid 6 months after the issue date.

Date: June, 18 2007

Subject: VCT JBOD (Disk-Array) Drive failure – VDARC/Apps Boot-up Problem.

Effectivity: All VCT GOC5 consoles with a Jarrell VDARC Chassis

Details:
A Software problem exists that incorrectly reports the location of the failing JBOD disk.

JBOD (Disk Array) Controller error message is reporting backwards> Adaptor 0 and Adaptor 1 are reverse creating a problem identifying the physical location of the defective disk drive. This failure mode can keep the DARC from booting up. This confusion could increase down time if the wrong disk drive is replaced.

Issue:
The message seen during the LSI logic test run during darc boot up. If a failure occurs, the test reports an error and hangs the darc boot up. Communication to the DARC will fail because the DARC does not bootup: Example-> ping or rsh darc will both fail.

The following Examples show different errors reported in various logs.

VDARC Console Window:
Using the VDARC Video cable connected or using telnet localhost 623 -> reset -> console

If the VCT Disk Array (JBOD) failure is causing the VDARC Node to stop the booting process, the user will be able to identify the failure. The console window will prompt the user to: Press any key to continue. The user can press any key on the keyboard to allow the VCT Disk Array to display the failure(s) and continue the VDARC Node boot process.

"Drive 2 on adaptor 0 is predicting future failure. Sense Qualifier is 00h. Press any key to continue...."

The above error is confusing because the actual failure is with Drive 2 on channel 1 (sd1)

After pressing any key, the darc finishes a LSI logic disk scan and boots Linux. The darc-gre-raid service *may* fail during Linux startup so darc /dev/md0 device won't be mounted and therefore CT Applications cannot be started.

Application boot-up failure pop-up Window

gre-verify-darc: failure - waiting for ping darc response

----

gesys-log message:

Tue Jun 5 11:51:52 2007 Error: 290014004
bay45 gre-verify-darc
gre-verify-darc.cpp 180
Unable to verify DARC system startup.

Operator Console Unix Window Messages:

Operator Console Apps Startup will attempt to recover. In the start-up (st) Unix shell the following ipmitool messages will appear and attempt to turn off and then on the VDARC Node for a recovery process, however, this recovery will fail.

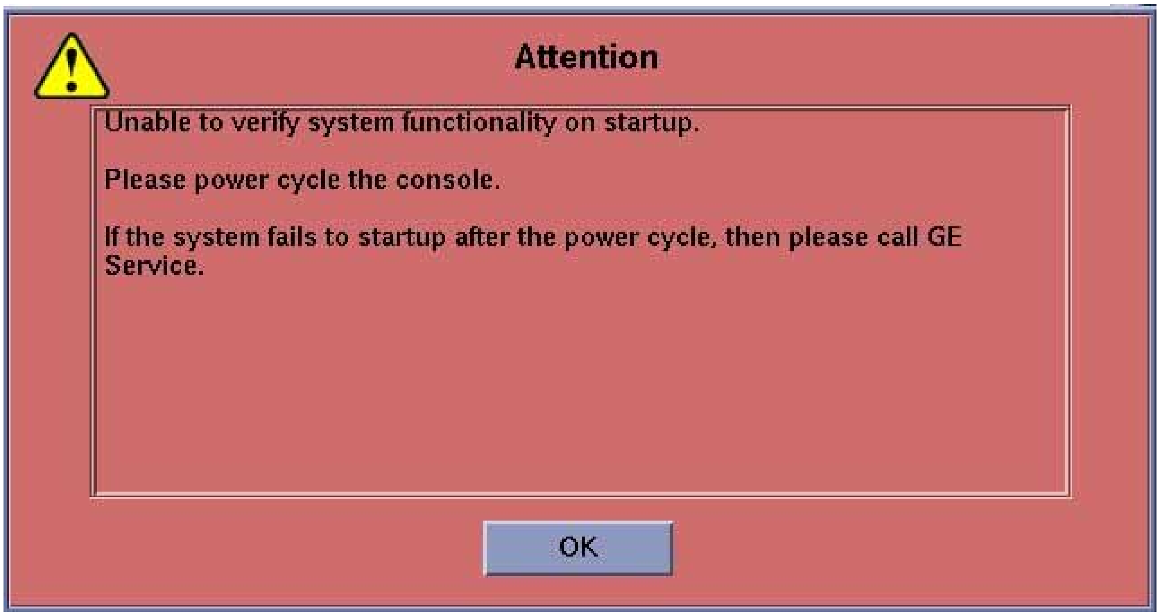
************* starting the LightSpeed system *************
Attentioninfo - ipmitool -H darc -P "" chassis power off
Chassis Power Control: Down/Off
info - ipmitool -H darc -P "" chassis power on
Chassis Power Control: Up/On

Resolution:
Run Smart Control (smartctl) commands to identify the failing drive. When error indicates adapter 0, this is actually SCSI channel 1 (sd1). See the following procedure on how to correctly identify the failing disk drive.

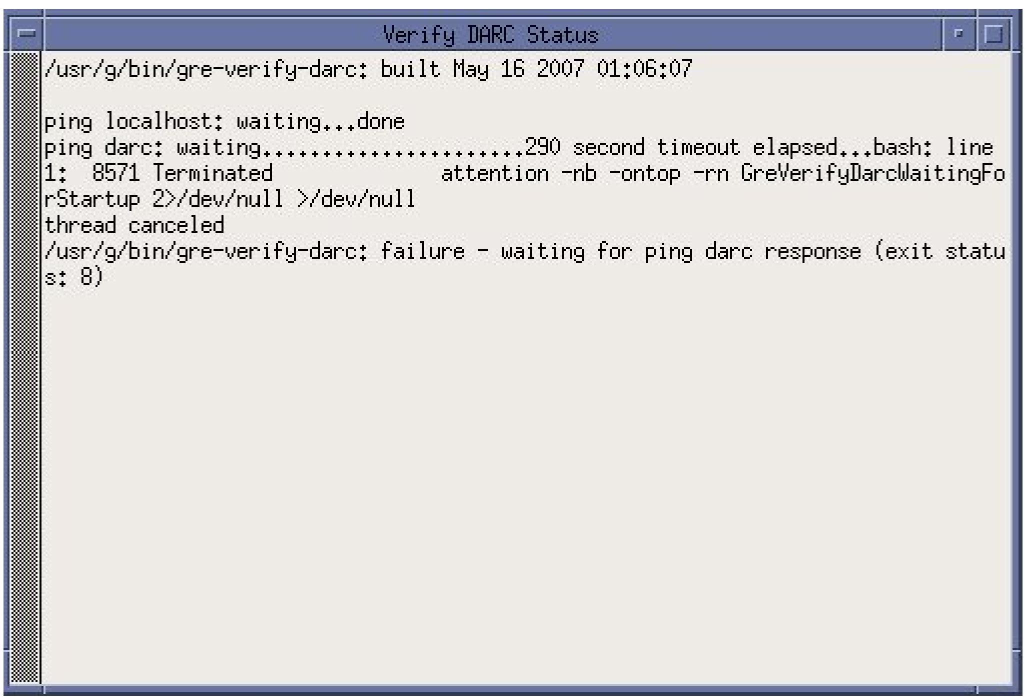
VCT JBOD (Disk-Array) Disk Drive Failure and/or Causing Application boot-up failures

1 VCT JBOD Smart Control Firmware Disk Issue

VDARC hang example: The GOC5 VCT Operator Console is turned on or rebooted. The command “ping darc” fails. When the Application Software is started the VDARC Node fails to boot.



The Attention Box will pop-up stating that the Host is “Unable to verify system functionality on startup.” When OK is selected the user may notice the following “Verify DARC Status” window. The gre-verify-darc process has failed because the VDARC cannot boot.



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```
*********** starting the LightSpeed system ***********
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Chassis Power Control: Down/Off
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Chassis Power Control: Up/On
```

If the gesyslog is accessed (example shown below), user will see the “Unable to verify DARC system startup” failure message:

```
{ctuser@hostname} cd  /usr/g/service/log
{ctuser@hostname} ls  ge*
gesys_bay45.log  gesys_bay45.log.05.22.2007.14.55.56.Z
{ctuser@hostname}] tail  -100f   gesys_bay45.log  &


1181044312      0       1       Tue Jun  5 11:51:52 2007        290014004      4
      bay45   gre-verify-darc
gre-verify-darc.cpp             180
Unable to verify DARC system startup.
/usr/g/bin/gre-verify-darc: failure - waiting for ping darc response (exit status: 8)
```

NOTE:

Example: To tail the gesys logs - perform the following:

{ctuser@hostname} **cd /usr/g/service/log**
{ctuser@hostname} **slay tail**
{ctuser@hostname} **ls ge***
g*esys_bsite’s log name.log gesys_site’s log name.log.05.22.2007.14.55.56.Z*
{ctuser@hostname} **tail -100f gesys_***site’s log name***.log &**

****** Use the latest uncompressed gesys.log shown! *******

1.1 Initiate console Telnet from Host to VDARC Node

Perform the telnet command as follows:

1. Verify another telnet window from the Host to the VDARC Node is not already open.
2. **Open a Unix Shell** -> at the Toolchest, with Application software down.
3. Perform the telnet command to the VDARC Node from the Host Computer:

{ctuser@hostname} **su -**

Password:**#bigguy**
[root@hostname] **service cliservice start**
dpcproxy server will be reported as already running or restarted
[root@hostname] **telnet localhost 623**
Trying 127.0.0.1...
Connected to localhost.
*** Indicates cliservice proxy server is running ***
Escape character is '^]'.
Server: **darc**
Username: **< Enter >**
Password: **< Enter >**
Login successful
dpccli> **reset**
OK
dpccli> **console**

1.2 Record the failure(s)

If the VCT Disk Array (JBOD) failure is causing the VDARC Node to stop the booting process, the user will be able to identify the failure.

The console window will prompt the user to: Press any key to continue. The user shall comply by pressing a key on the keyboard to allow the VCT Disk Array to display the failure(s) and continue the VDARC Node boot process.

1. Record the failure(s). An example failure for a Disk in the Disk Array is provided below.
2. Press any key on keyboard to continue the VDARC Node boot process.

This is an example of the firmware smart control error you may see: "Drive 1 on adaptor 1 is predicting future failure. Sense Qualifier is 00h. Press any key to continue...."

1.3 Utilizing SMART Control Commands to Identify the Failing drive(s).

Perform the telnet command as follows:

1. **Open a Unix Shell** -> At the Toolchest, with Application software down.
2. Verify another telnet window from the Host to the VDARC Node is not already open.
3. Verify the "SMART Health Status output for all Disk Array Drives present (sda thru sdh) using the 'smart control' command. Open a Unix Shell and type the following:

NOTE: Perform the following command for each Disk in the Disk Array to determine which drive(s) is faulty. Replace all failing drives.

a. {ctuser@hostname} **su -**
b. Password: **#bigguy**
c. [root@hostname] **rsh darc**

Last login: Tue Jun 5 12:12:07 from oc

d. [root@darc] **smartctl -a /dev/sda**

SMART Health Status:

e. [root@darc] **smartctl -a /dev/sdb**

SMART Health Status:

f. [root@darc] **smartctl -a /dev/sdc**

SMART Health Status:

g. [root@darc] **smartctl -a /dev/sdd**

SMART Health Status:

h. [root@darc] **smartctl -a /dev/sde**

SMART Health Status:

i. [root@darc] **smartctl -a /dev/sdf**

SMART Health Status:

j. [root@darc] **smartctl -a /dev/sdg**

SMART Health Status:

k. [root@darc] **smartctl -a /dev/sdh**

SMART Health Status:

l. [root@darc] **exit**

1.4 Understand the SMART Control output

An example of a failing drive is provided below. Verify the "SMART Health Status" output:

```
[root@darc ~]# smartctl -a /dev/sda

smartctl version 5.33 [i386-redhat-linux-gnu] Copyright (C) 2002-4 Bruce Allen

Home page is http://smartmontools.sourceforge.net/

Device: SEAGATE  ST373454LC       Version: 0002
Serial number: 3KP42LBG000097177EB5
Device type: disk
Transport protocol: Parallel SCSI (SPI-4)
Local Time is: Tue Jun  5 12:28:29 2007 CDT
Device supports SMART and is Enabled
Temperature Warning Enabled

SMART Health Status: FAILURE PREDICTION THRESHOLD EXCEEDED
[asc=5d,ascq=0]

Current Drive Temperature:     34 C
Drive Trip Temperature:        68 C
Vendor (Seagate) cache information
  Blocks sent to initiator = 98388816
  Blocks received from initiator = 396302262
  Blocks read from cache and sent to initiator = 44535918
  Number of read and write commands whose size <= segment size = 1773227
  Number of read and write commands whose size > segment size = 12370

Error counter log:

         Errors Corrected by          Total   Correction     Gigabytes    Tot al
            EEC          rereads/    errors   algorithm      processed    unc orrected
```

```
        fast | delayed   rewrites  corrected  invocations   [10^9 bytes]  err ors
read:     165662      15        8    165685    1184447       116.801   0
write:        0        0        0        0         0       195.450   0
Non-medium error count:       62
Error Events logging not supported
[GLTSD (Global Logging Target Save Disable) set. Enable Save with '-S on']

SMART Self-test log

Num  Test            Status              segment  LifeTime  LBA_first_err [ SK ASC ASQ]
    Description                        number   (hours)
# 1  Background long   Failed in segment -->       -   960  0x        6c32368 [ 0x3 0x11 0x0]
# 2  Background short  Completed                  -   960                - [ -   -    -]
Long (extended) Self Test duration: 1150 seconds [19.2 minutes]
```

An example of a passing drive is provided below. Verify the "SMART Health Status" output:

```
[root@darc ~]# smartctl -a /dev/sdb
smartctl version 5.33 [i386-redhat-linux-gnu] Copyright (C) 2002-4 Bruce Allen
Home page is http://smartmontools.sourceforge.net/

Device: SEAGATE  ST373454LC       Version: 0002
Serial number: 3KP3NQW0000097089GZ9
Device type: disk
Transport protocol: Parallel SCSI (SPI-4)
Local Time is: Tue Jun  5 12:28:44 2007 CDT
Device supports SMART and is Enabled
Temperature Warning Enabled
SMART Health Status: OK

Current Drive Temperature:     40 C
Drive Trip Temperature:        68 C
Vendor (Seagate) cache information
  Blocks sent to initiator = 988796
  Blocks received from initiator = 45181853
  Blocks read from cache and sent to initiator = 182708
  Number of read and write commands whose size <= segment size = 78535
  Number of read and write commands whose size > segment size = 194

Error counter log:
           Errors Corrected by           Total   Correction     Gigabytes    Total
               EEC          rereads/    errors   algorithm      processed    uncorrected
```

```
          fast | delayed   rewrites  corrected  invocations   [10^9 bytes]  errors
read:       1647        0         0      1647       1647          1.462          0
write:         0        0         0         0          0         58.052          0

Non-medium error count:        0

Error Events logging not supported

[GLTSD (Global Logging Target Save Disable) set. Enable Save with '-S on']
No self-tests have been logged
Long (extended) Self Test duration: 1150 seconds [19.2 minutes]
[root@darc ~]#
```

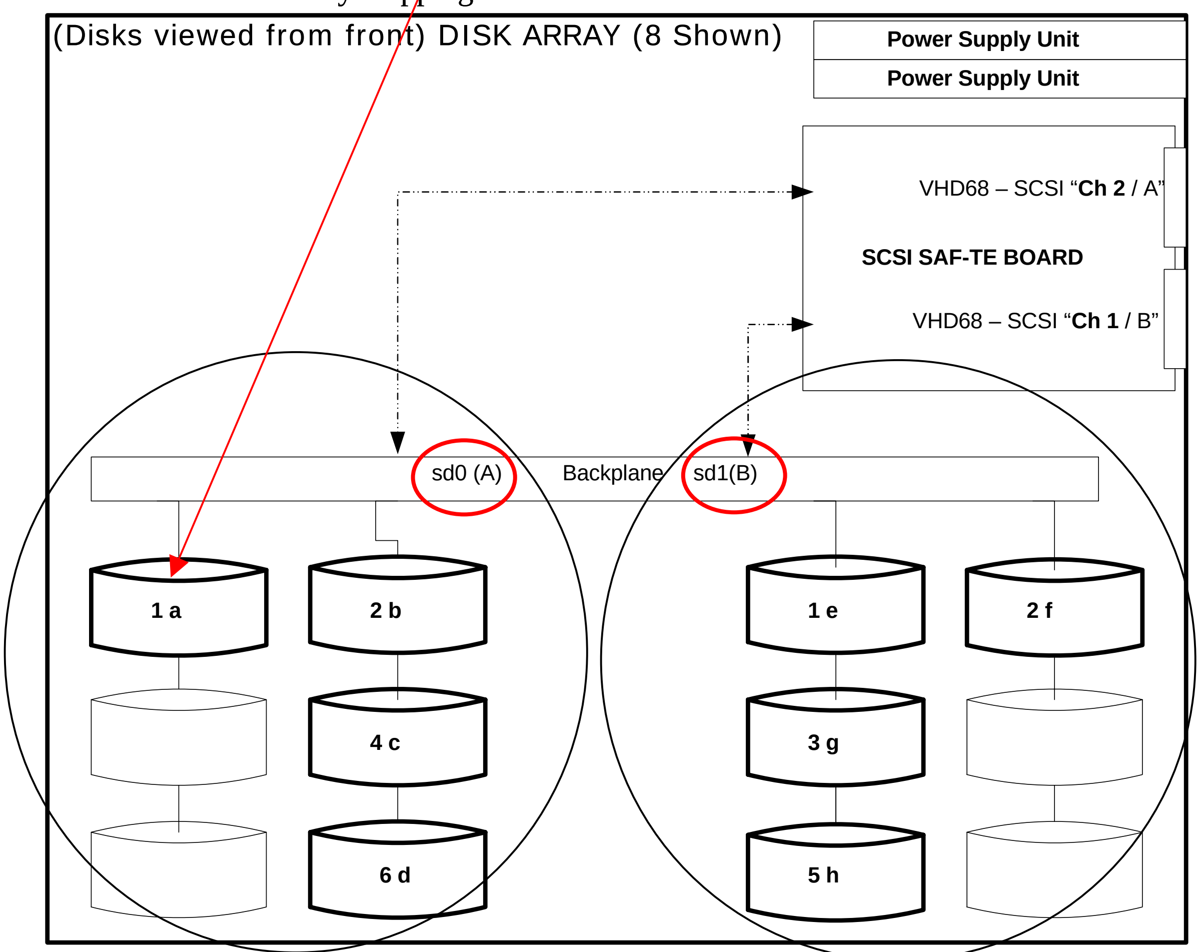
1.5 Identify the failure location(s)

The Smart Control **(smartctl)** command correctly Identifies the failing drive. Remember the following DARC console window is confusing **"Drive 1 on adaptor 1 is predicting future failure. Sense Qualifier is 00h. Press any key to continue...."**

Use the Smart Control (smartctl) command to correctly Identifies the failing drive.
In the example from section 1.4 dev/sda drive reports defective with the Smart Control command.

Currently several drives have been identified with the following failure message:
FAILURE PREDICTION THRESHOLD EXCEEDED [asc=5d,ascq=0]
unrecovered read error: "ASC=11 ASCQ=00, SelfTestByte=00, VendorSpecificByte=0F"

Documented Disk Array mapping:



1.6 Working Around a Disk Failure

If one or two disk drives in the Disk Array are defective, the user may be able to continue running until the replacement drive/s arrive, however, all patient data maybe lost.

The VCT Disk Array (JBOD) will allow a total of two (2) disks to be dropped (from 8 to 6 drives). Three (3) disks must be good on each side of the Disk Array for Application Software to boot up and scan. If a total of seven (7) disks are good and one (1) disk is bad, only six (6) disks will be used. The JBOD disk array can only have one defective drive per side. The JBOD operates in a symmetrical fashion if on one drive is defective a good disk from the other side will be dropped out until a good replacement disk is inserted and the replacement procedure is followed.

Working Around a Disk Failure:

1. Pull the first bad disk out of the Disk Array (JBOD).
2. If a second bad disk is present, verify it is pulled out of the other side of the Disk Array.
3. Do not insert any bad disks into the Disk Array.
4. As long as only one (1) bad disk is removed on each side of the (JBOD) the VDARC will successfully boot up and Application Software will startup. A message will be present in the gesyslog stating that 6 disks are present. The Customer may be informed by a pop-up window as well.

1.7 Order Disk Replacement(s) immediately

The VCT Operator Console is designed to run with eight (8) disks presents {it will run with 6 disks, 3 good disks on each controller}. Order replacement Disks immediately.

1.8 Reboot to verify the VDARC Node will boot

Perform a reboot command with Application software down. Verify the VDARC Node boots up without user input. Verify that the VCT System can scan properly.

1.9 Replacement

Refer to the LightSpeed Family for 7.x, select the Replacement Tab, select the Console tab, select Disk Array (JBOD) disk FRU Replacements. Perform all required tests.

1.10 Regen the database in reconfig

With Application Software down, open a Unix Shell. Become root and perform the reconfig command. Select REGEN DATABASE and Accept. The Operator Console will reboot.

2 Finalization

1. Perform sanity scans (e.g., scout, axial and helical).

NOTE: If the System will not scan or reconstruct, verify the memory on the VDARC and VIG Node(s) is correct (lhinv). Verify the VDIP (no loopback) diagnostics pass [root@darc] vdip_menu –A). Verify the VRAC diagnostics pass ([root@ig#] vrac_menu –a). Verify the System Error Log report gre-raid success with 8 drives present with application software up. Refer to Console Diagnostics tool, Troubleshooting tab or Replacement tab for further assistance.

2. Verify the image(s) reconstruct and are properly displayed.
3. Install the front and rear Operator Console covers.

Affected Service Publications:

- 5193575-800 - LS 7.X Limited Access Service Info (Adv), Rev 2
- 5178786-800 - Discovery VCT PET-LS7.X Service Methods (Adv), Rev 2

PQR/SPR #: FCTge 29467

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